Coherence sweet spots are frequency minima in disguise: Metric sensitivity in transmon and fluxonium design spaces

Ishita Bera[1⋆]

1 Independent researcher

⋆ ishitabera4@gmail.com

Abstract

At fixed quality factor, dielectric loss makes a lower-frequency transmon-like qubit relax more slowly. This has a stronger consequence for design studies than is usually acknowledged. Under a common noise model, $\log T_2$ over the transmon (E_J, E_C) plane correlates with $\log f_{01}$ at $r = -0.9999$ within the transmon regime, and over the fluxonium (E_J, E_L) plane at $r = -0.9878$. Because modeled T_2 on each plane is largely a function of the transition frequency, a raw T_2 maximum primarily identifies the lowest-frequency corner of the swept domain rather than an optimal design. Both our initial headline optima lie at grid edges outside the usual bands.

Controlling for frequency changes these conclusions. For the transmon, we find no interior optimum in E_J/E_C at fixed f_{01}: T_2 varies by only 3.2% over $E_J/E_C = 11$–118, while the anharmonicity decreases by a factor of 7.3. Charge insensitivity is therefore bought with anharmonicity rather than coherence, and any reported optimum depends on the imposed regime boundary.

Within this model, the fluxonium has an interior optimum once second-order $1/f$ flux dephasing is included, but at only 5.4–30 MHz, a convention-dependent location, where $\bar{n}_{\mathrm{th}} = 10$–$57$ requires active initialization. Across architectures there is no single answer: the fluxonium/transmon ratio spans ∼400× across the metrics we consider, and at matched frequency the transmon is ahead, with a median T_2 ratio of 0.37× at 1 GHz under our noise priors. A lab-frame pulse simulation puts the coherence-to-gate-time ratio at a model-dependent 17–27×, between the two heuristic estimates, partly because the fluxonium drive couples 10.2× more strongly to its leakage transition than to its qubit transition. Critical-current noise, read as fractional rather than as `scqubits` applies it, dominates the fluxonium.

Contents

1 Introduction **2**

2 Related Work and Contribution **4**
2.1 What is already established 4
2.2 Contribution 4
2.3 A limitation of the present comparison 5

3 Model and Conventions **6**
3.1 Noise channels 6
3.2 Second-order flux dephasing 7
3.3 A declared modeling choice 8

4 Validation **8**

5 The Transmon Design Plane **9**
5.1 The confound 9
5.2 No interior optimum in this model 9

6 The Fluxonium Design Plane **10**
6.1 An interior optimum, created by second-order dephasing 11

7 Dominant Noise Channels **12**

8 The Cross-Architecture Comparison **13**
8.1 Bias robustness, the one unambiguous asymmetry 13
8.2 No single ratio 14
8.3 The raw advantage is a frequency effect 14
8.4 Estimating gate durations by simulation 15

9 Calibration Against Measurement **16**

10 Limitations **18**

11 Conclusion **18**

12 Outlook **19**

Code and data availability **19**

References **20**

1 Introduction

Superconducting qubit design is routinely discussed in terms of *sweet spots*: bias points at which the transition frequency is stationary with respect to a noise coordinate, so that first-order dephasing vanishes. The transmon has charge sweet spots at integer and half-integer offset charge, and its exponentially suppressed charge dispersion keeps it nearly insensitive in between [1]; the fluxonium achieves it at half a flux quantum, where $\partial\omega_{01}/\partial\Phi = 0$ by symmetry [2]. A natural extension carries the idea from bias to the circuit parameters themselves: the point of best modeled T_2 over a swept parameter plane is read as an optimal design, a coherence sweet spot in *design* space. The term is ours, introduced to name that reading.

This paper argues that this extension does not survive scrutiny: under the noise models studied, it systematically selects for low transition frequency rather than for a design. We address three questions:

Q1 Where are the coherence optima over the transmon (E_J, E_C) plane, and over the fluxonium (E_J, E_L) plane, selected (E_C, E_L) slices and flux-bias sweeps?

Q2 Which noise channel dominates where?

Q3 Does the fluxonium's flux sweet spot buy more robustness than the transmon's charge insensitivity, and at what cost?

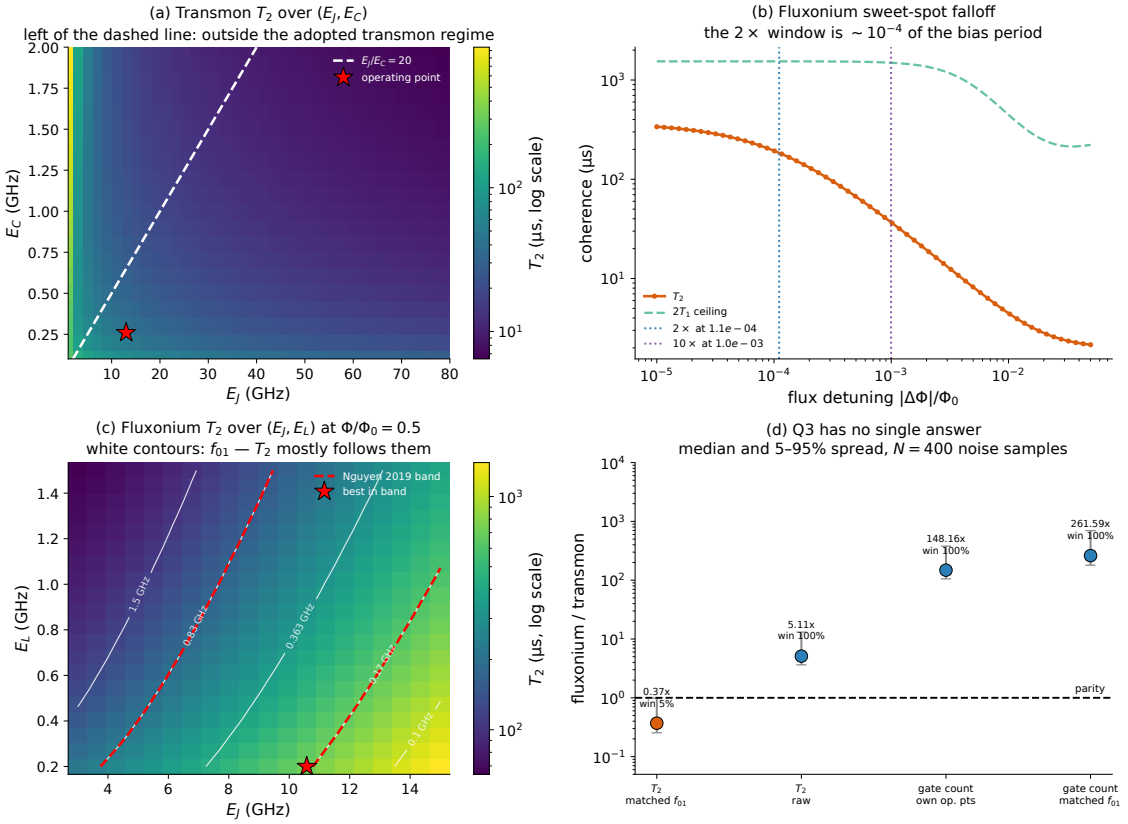


Figure 1: The study in four panels. (a) Transmon T_2 over (E_J, E_C) at $n_g = 0$; left of the dashed line ($E_J/E_C < 20$) lies outside the transmon regime adopted here. Points there are not used for any transmon headline result; the fixed-frequency slice of Section 5 extends to $E_J/E_C = 11.4$ only to show the trend. (b) The fluxonium sweet-spot falloff on a log-detuning axis, with the $2T_1$ ceiling and the two- and tenfold loss points; on a linear flux axis the entire collapse hides inside the first 0.1% of the plot. (c) Fluxonium T_2 over (E_J, E_L) at half flux, with f_{01} contours in white — the T_2 structure mostly follows them (12% rms scatter at fixed f_{01}), which is the confound this paper is about. (d) The Q3 answer under each metric, with the 5–95% spread over $N = 400$ noise-amplitude draws; orange marks the metrics in which the fluxonium wins fewer than 90% of samples. These Monte Carlo medians span ∼700× because they include the gate count at matched frequency, whose matched transmons lie outside the design envelope; the ∼400× quoted in the text is the deterministic span across the metrics of Table 2 (242× against 0.62×).

Our central methodological claim is that Q1 and Q3 cannot be answered without controlling for the transition frequency, and that once it is controlled the answers our model gives to both change sign or dissolve. Our central negative result is that at matched frequency the fluxonium has no coherence advantage in this model: the transmon is ahead at every matched frequency we test, and in 95% of noise draws.

Figure 1 summarizes the study. All results are model-based estimates under literature noise amplitudes. They are not predictions of device performance, and Section 9 quantifies how far they sit from measurement.

2 Related Work and Contribution

2.1 What is already established

The frequency scaling on which we build is not a new idea. Dielectric loss at surfaces and interfaces, commonly attributed to two-level-system defects, is a major relaxation mechanism in superconducting qubits; in 3D transmons the relaxation rate scales approximately with the surface participation of lossy dielectrics [3]. For a weakly anharmonic qubit at fixed effective quality factor and low temperature ($hf_{01} \gg k_B T$), $\Gamma_1 \approx \omega_{01}/Q$: the charge matrix element, the charging energy and the transition frequency are tied together by the harmonic-oscillator relations, so a lower-frequency qubit relaxes more slowly. The fluxonium's matrix elements do not follow those relations. At finite temperature upward transitions add a further factor $\coth(hf_{01}/2k_B T)$, which is 1.00 at 4.92 GHz but 1.91 at 0.363 GHz and 8.4 at 75 MHz at 15 mK, so the simple law fails in the low-frequency fluxonium regime. Design-space optimization under explicit noise models is likewise established: Papič *et al.* identify optimal transmon parameter ranges using comprehensive noise modeling that includes charge-parity effects [4], while Ram *et al.* present a staged framework spanning circuit design, spectral analysis, and calibration for both architectures [5]. Experimentally, fluxonium coherence has advanced well beyond the devices against which we calibrate our model: Somoroff *et al.* report $T_2^* = 1.48\,\mathrm{ms}$ together with an average gate fidelity of 0.99991 [6].

Two works are sufficiently close to the present study to warrant more precise comparison. McRae *et al.* [7] argue that coherence metrics cannot yet be reliably compared across measurements, devices, and laboratories because temporal fluctuations, TLS dynamics, and incompletely reported experimental conditions can make the claimed differences unreliable. Their concern is experimental reproducibility. The present paper develops a modeling analog of that argument: a comparison extracted from a swept design space is likewise unreliable unless both the controlled variable and the figure of merit are specified alongside the reported value.

Lange *et al.* [8] directly compare transmon and fluxonium lattices and conclude that fluxonium architectures may show lower cross-talk when designed for similar gate speeds and fidelities, a matching they argue from coupling strengths in published two-qubit-gate experiments. Their figures of merit are residual ZZ cross-talk and state localization rather than coherence. Our work is complementary: it applies the same principle of controlled comparison to coherence and asks how the conclusion changes when transition frequency is held fixed.

2.2 Contribution

Our contribution is not the frequency scaling itself, but its consequence for how design studies are interpreted:

1. We quantify the frequency confound in both architectures, finding $|r| > 0.98$ between modeled $\log T_2$ and $\log f_{01}$ across the relevant design planes. As a result, a raw T_2 optimum can primarily reflect the lowest-frequency boundary of the sweep rather than an intrinsic design optimum.

2. We show that the transmon's anharmonicity-based gate-count optimum within the 3–6 GHz band is pinned to the imposed regime threshold across eight thresholds spanning $E_J/E_C = 15$–60. The substantive choice is therefore the threshold itself, rather than the subsequent optimization within that domain.

3. We show that including second-order $1/f$ flux dephasing produces an interior fluxonium optimum that is absent from a first-order treatment.

4. We show that the transmon–fluxonium coherence comparison varies substantially across defensible figures of merit, and we estimate the gate-count comparison, the one most sensitive to its assumptions, at 17–27× with a lab-frame pulse simulation.

We are not aware of prior work that explicitly examines how strongly the transmon–fluxonium coherence comparison depends on the chosen metric. This rests on the survey above, which is not exhaustive; the individual ingredients are known, and the claim concerns their combination. Our contribution is to show this dependence and how controlling for transition frequency changes the resulting design conclusions.

2.3 A limitation of the present comparison

Matching f_{01} removes the dominant confound but not all of it. Dielectric loss also depends on geometry and surface participation, which differ between architectures and which a Hamiltonian-level model does not capture [3], and real devices fluctuate in ways no static model represents [7]. Frequency matching is necessary to separate architecture-dependent effects from the transition-frequency difference, but not sufficient for a complete hardware comparison; comparing devices at their intended operating points answers a different, also legitimate, question. Our matched-frequency numbers should be read as removing one large confound rather than as a fully controlled experiment.

A second limitation follows from the matching itself. The fluxonium's sweet-spot frequency lies below the transmon band, so the matched transmons sit at 0.5–2 GHz with $E_J/E_C = 325$–5050 and $|\alpha| \approx 10$ MHz, far outside the design envelope adopted in Section 5. The matched transmon at each frequency is the one with the highest T_2 allowed by $E_J/E_C \geq 50$, and since modeled T_2 rises slowly with E_J/E_C that is the most favorable transmon the scan contains, at the edge of its E_C range. The choice barely matters for T_2: holding the matched transmons at $E_J/E_C = 50$ instead changes their T_2 by at most 1.3% and the 1 GHz Monte Carlo median of Section 8 from 0.37× to 0.38×. It does mean the matched transmons lie outside the adopted design envelope and are unsuited to the assumed gate speeds, and gate-based metrics at matched frequency should not be read as device comparisons.

A third limitation is that the two noise models are not symmetric. Quasiparticle tunneling is modeled for the fluxonium but not for the transmon, for which `scqubits` provides no such channel; inductive loss in the fluxonium's superinductance, a known loss channel in real devices, is omitted; and only the fluxonium model is calibrated against measurement (Section 9). The quasiparticle asymmetry is not harmless. Evaluating the same `scqubits` quasiparticle spectral density with the transmon's $|\langle 0|\sin(\varphi/2)|1\rangle|^2$ gives $T_1^{\mathrm{qp}} \approx 5.6\,\mu$s at the transmon operating point for the default $x_{qp} = 3\times10^{-6}$, shorter than its dielectric T_1 of 32 µs; at $x_{qp} = 10^{-8}$ it is 1.7 ms, a 2% correction. The leading-order analytic transmon result of Catelani *et al.* [9] gives 5.55 µs at the same density, within 1.5%. Quasiparticle tunneling changes charge parity, and the extended-phase matrix element used here neglects the difference between the two parity sectors. That is an approximation: at $E_J/E_C \geq 20$ parity-dependent spectral differences are exponentially suppressed, which is why the two estimates agree, but it does not make every effect of quasiparticle tunneling identical in the two sectors. For the modeled quasiparticle-induced $0 \to 1$ relaxation, omitting the channel is approximately equivalent to taking a small transmon quasiparticle density, consistent with the bound Nguyen *et al.* infer for fluxonium. It does not exclude other quasiparticle- or charge-parity-related errors [4]. The fluxonium, meanwhile, carries the 300× higher default. That default is irrelevant at the flux sweet spot, where the fluxonium term vanishes, but our transmon numbers are conditional on a low x_{qp}.

The other two asymmetries can bias the cross-architecture ratios, in opposite directions, by amounts we do not quantify.

3 Model and Conventions

3.1 Noise channels

Coherence is estimated using `scqubits` [10]. Independent relaxation and dephasing rates are combined as

$$\frac{1}{T_2} = \frac{1}{2T_1} + \sum_k \frac{1}{T_{\phi,k}}, \tag{1}$$

with noise channels selected separately for each architecture. The fixed-frequency transmon considered here contains no flux-tunable loop, so $1/f$ flux dephasing is not included. In the fluxonium model considered here, the inductive shunt removes the dependence on offset charge entirely, so charge dephasing is not included. Table 1 summarizes the channel assignments.

Throughout, E_J, E_C and E_L are quoted as frequencies (E/h, in GHz), following `scqubits`, and the anharmonicity is $\alpha = f_{12} - f_{01}$. T_1 is the inverse of the total relaxation rate, the sum of the downward rate and the thermally activated upward rate ($\Gamma_\downarrow + \Gamma_\uparrow$), which is the `scqubits` default. Each $1/f$ channel has spectral density $S_\lambda(\omega) = 2\pi A_\lambda^2/|\omega|$ in the noise coordinate λ, the `scqubits` convention, which Nguyen *et al.* also use for Eq. (2); we use the package's default cutoffs and measurement time, and the first-order dephasing rate depends explicitly only on the infrared cutoff $\omega_{\text{low}} = 2\pi \times 1\,\text{Hz}$ and the measurement time $t_{\text{exp}} = 10\,\mu\text{s}$. A_Φ is in units of Φ_0. $A_{n_g} = 10^{-4}$ is defined as an amplitude in the dimensionless coordinate n_g, which counts offset charge in units of the Cooper-pair charge $2e$; it corresponds to $2 \times 10^{-4} e$. This is Koch *et al.*'s convention: their worked example uses $A = 10^{-4}$ with $\partial E_{01}/\partial n_g$ in Eq. (5.9) [1], and Section 4 reproduces it. It is also what `scqubits` computes, although the package labels the default as in units of e. Critical-current noise is where we depart from the package. `scqubits` documents A_{cc} in units of the critical current, but applies it to the dimensionless operator $\partial H/\partial E_J = -\cos\varphi$, so the amplitude acts as an absolute fluctuation of E_J in GHz. We read A_{cc} as the fractional fluctuation $\epsilon_I = \delta I_c/I_c$, as the literature amplitudes it stands for are: since $E_J = \hbar I_c/2e$, the perturbation is $\delta H(t) = -E_J \cos\hat{\varphi}\, \epsilon_I(t)$ with $S_{\epsilon_I}(\omega) = 2\pi A_{\text{cc}}^2/|\omega|$, and we pass $A_{\text{cc}} E_J$ to the package. The two readings differ by E_J/h in GHz: 8.9× for the nominal fluxonium and 13× for the transmon operating point. Under the absolute reading the critical-current rate of a transmon at fixed f_{01} would also depend on E_J/E_C: since $\partial f_{01}/\partial E_J \approx \sqrt{2E_C/E_J}$, it would fall as $(E_J/E_C)^{-1/2}$, the dephasing time growing as $\sqrt{E_J/E_C}$. A fractional E_J fluctuation instead shifts f_{01} by $E_J\, \partial f_{01}/\partial E_J \approx (f_{01} + E_C)/2$ at any E_J/E_C. The choice matters. Under the absolute reading, which the package applies by default, the nominal fluxonium T_2 would be 1130 µs rather than 369 µs, the raw fluxonium/transmon ratio 18.1× rather than 8.4×, the matched-frequency ratios 1.07–2.16× rather than 0.62–0.82×, and the calibration of Section 9 2.6× optimistic on T_2 rather than 1.5×. Critical-current dephasing carries a median 76% of the fluxonium's total decoherence rate over its design plane, so its amplitude is the dominant uncertainty for that architecture. Koch *et al.* use the same fractional amplitude, $\bar{A} = A/I_c$, independent of E_J [1]. The convention and the value are separate questions. Measured $1/f$ critical-current noise in Al/AlOx and Nb-based junctions [11–13], extrapolated to 15–20 mK and junction areas of 0.01–0.1 µm², gives $\sqrt{S_{\delta I_c/I_c}(1\,\text{Hz})} \approx 1.4 \times 10^{-7}$–$1.4 \times 10^{-6}\,\text{Hz}^{-1/2}$, clustering near 2–6 × $10^{-7}\,\text{Hz}^{-1/2}$. For a $1/f$ spectrum this square root at 1 Hz is the dimensionless coefficient A_{cc} of our convention, up to the factor $\sqrt{2}$ between one- and two-sided spectra, which the measurements do not specify; the temperature scaling

Table 1: Noise channels by architecture.

Channel	`scqubits` method	Transmon	Fluxonium
Dielectric loss (T_1)	`t1_capacitive`	✓	✓
Quasiparticle (T_1)	`t1_quasiparticle_tunneling`	–	✓
$1/f$ flux, 1st order	`tphi_1_over_f_flux`	–	✓
$1/f$ flux, 2nd order	*this work*	–	✓
Charge / offset	`tphi_1_over_f_ng`	✓	–
Critical current	`tphi_1_over_f_cc`	✓	✓

below ∼0.3 K is not measured, and the extrapolations disagree by about 2.5×. Our baseline $A_{\mathrm{cc}} = 10^{-7}$ therefore lies near the optimistic end of the range extrapolated from published measurements, Koch *et al.* use 10^{-6}, and the Monte Carlo range of Section 8, 10^{-7}–10^{-6}, spans the extrapolations. The adopted amplitude is a literature-motivated modeling assumption, not a device-specific measurement.

Equation (1) is an approximation: we represent each relaxation and dephasing contribution by an effective exponential rate and add the rates to define a comparative T_2. Independence of the noise processes, exponential decay and additivity are separate assumptions, and the construction does not in general reproduce the decay envelope. Linear coupling to quasistatic $1/f$ noise instead produces approximately Gaussian dephasing, and at a sweet spot, where the coupling is quadratic, the accumulated phase is quadratic in the noise and the envelope differs from both forms. The first- and second-order flux terms also arise from the same noise process, so adding their rates is a further approximation, most relevant at small detunings where the two are comparable. Equation (1) therefore summarizes a non-exponential decay by a single effective time. We use it, as is standard in design studies, to compare designs rather than to predict measured decay curves.

We use $Q_{\mathrm{cap}} = 10^6$, $A_{n_g} = 10^{-4}$, $A_\Phi = 10^{-6}\,\Phi_0$, and $A_{\mathrm{cc}} = 10^{-7}$. The chosen flux-noise amplitude, $A_\Phi = 1\,\mu\Phi_0$, is slightly below the values measured on flux-sensitive devices, $1.4\,\mu\Phi_0$ [14] and $1.8\,\mu\Phi_0$ [2], and is therefore optimistic. Temperature, quasiparticle density, and the $1/f$ cutoffs are assumed parameters, taken from the `scqubits` defaults ($T = 15\,\mathrm{mK}$, $x_{\mathrm{qp}} = 3 \times 10^{-6}$, $t_{\mathrm{exp}} = 10\,\mu\mathrm{s}$) and are recorded with every result because these parameters materially affect the predicted coherence times.

3.2 Second-order flux dephasing

`scqubits` evaluates $1/f$ dephasing to first order in the sensitivity $\partial\omega_{01}/\partial\lambda$. At an exact flux sweet spot, $\partial\omega_{01}/\partial\Phi = 0$, so the first-order dephasing rate vanishes and the corresponding calculated dephasing time diverges [10]. The leading flux sensitivity at half flux is therefore second order. We include the second-order contribution using the expression employed for fluxonium by Nguyen *et al.* [2], following Ithier *et al.* [15]:

$$\Gamma_\lambda^{(2)} = \left| \frac{\partial^2 \omega_{01}}{\partial \lambda^2} \right| A_\lambda^2, \tag{2}$$

where λ is the dimensionless noise coordinate ($\lambda = \Phi/\Phi_0$ for flux, $\lambda = n_g$ for charge), so the curvature is in $\mathrm{rad\,s^{-1}}$ and A_λ is dimensionless: $A_\lambda = 10^{-6}$ for flux is $A_\Phi = 1\,\mu\Phi_0$. The absolute value keeps the rate positive at a sweet spot of either curvature. Equation (2) contains no logarithmic factor under this convention, and it is a phenomenological effective rate rather than a derivation of the decay envelope. Alternative treatments introduce factors involving $|\ln(\omega_{\mathrm{low}} t_{\mathrm{exp}})|$, producing substantial quantitative changes in the inferred dephasing time.

Rather than selecting among these conventions implicitly, we report the associated sensitivity explicitly (Section 10). The transmon at $n_g = 0$ is also a first-order sweet spot, for charge, so the same divergence occurs there. The analogous second-order charge term, evaluated with Eq. (2) at $\lambda = n_g$ and A_{n_g}, gives $T_\phi \geq 0.15$ s everywhere with $E_J/E_C \geq 20$ (78 s at the operating point), and we omit it as negligible. That estimate uses the same convention as the flux term but, unlike it, has not been checked against an independent published number. It is negligible throughout the adopted $E_J/E_C \geq 20$ regime under our assumed noise model, and we do not rely on that omission when interpreting behavior below the regime boundary. The required curvature is evaluated using Richardson extrapolation, since a simple central difference remains 4–10% from its converged value in the deepest-well part of the scan ($E_J = 40$–45 GHz, $E_L = 0.2$ GHz), though the finite-difference estimates agree to within 0.2% under the step-size extrapolation at the interior optima of Section 6.

3.3 A declared modeling choice

Passing Q_{cap} explicitly replaces the `scqubits` default frequency-dependent quality factor, $Q_{\text{cap}}(\omega) \propto \omega^{-0.7}$ [16], with a constant value. At the fluxonium sweet spot, these two assumptions change the calculated T_1 by a factor of 7.1, making this the largest single modeling choice in the study. The resulting uncertainty is not symmetric: Nguyen *et al.* fit their measured T_1-versus-frequency dependence with $\tan\delta_C \propto \omega^{0.15}$, and at $f_{01} = 0.363$ GHz the constant-Q model lies 1.52× from their measured scaling, compared with 4.7× for the `scqubits` default. That exponent was fit jointly with thermal effects and is not an independent measurement of the loss tangent, but it favors the constant-Q_{cap} model, which we therefore retain as the baseline because it is closer to the fitted frequency scaling of those devices over the range examined here, while stating the assumption explicitly. The frequency confound of Sections 5 and 6 does not depend on this choice. Under the frequency-dependent default, $\text{corr}(\log T_2, \log f_{01})$ is −0.9997 for the transmon ($E_J/E_C \geq 20$) and −0.9718 for the fluxonium, against −0.9999 and −0.9878 here, and both raw maxima remain at the lowest-frequency grid point. The cross-architecture comparison does depend on it. Under the default, the raw T_2 ratio of Section 8 rises from 8.4× to 9.6×, and the deterministic matched-frequency ratios fall from 0.82, 0.62 and 0.62× at 0.5, 1 and 2 GHz to 0.37, 0.37 and 0.48×: the longer dielectric T_1 leaves the fluxonium's critical-current dephasing a larger share of its rate budget (91% rather than 75% at 1 GHz). The size of the matched-frequency ratio therefore depends on the loss model, although under both it favors the transmon.

4 Validation

Forty-four checks of the numerical pipeline run before the results presented below are generated. They are numerical checks, analytic benchmarks and selected experimental comparisons, and they test the numerics and the stated limits rather than the completeness of the noise model. They include exact Hamiltonian symmetries to machine precision, basis-size convergence, analytic limits from Koch *et al.* [1], and, in several cases, comparison with published data from experimental devices.

The strongest analytic benchmark is Koch *et al.*'s worked charge-dephasing example. For $E_J = 30$ GHz, $E_C = 0.35$ GHz, and $A = 10^{-4}$, they report $T_2 \sim 8$ s. Substituting our computed charge dispersion into their Eq. (5.9) gives $T_2 = 8.75$ s. At $E_J/E_C = 50$, the maximum charge-dispersion derivative is 0.830 times their asymptotic value of $1.5 \times 10^{-4} E_C$, consistent with the asymptotic expression being approached from below.

Against Manucharyan *et al.* [17], the lowest two transitions at $\Phi_{\text{ext}} = 0$ are reproduced to better than 0.7% for each of the paper's three published parameter sets. A single four-

parameter fit additionally reproduces all six digitized transition lines across two flux biases with an RMS error of 0.0102 GHz. The three published parameter sets—the rounded main-text values and two Table I least-squares fits—produce a 9.5% spread in f_{01} at the flux sweet spot. Agreement quoted only against the rounded main-text parameters would therefore overstate the experimental constraint.

The third transition at $\Phi_{\text{ext}} = 0$ is excluded from this comparison for a physical rather than numerical reason. Its bare frequency lies only 0.14–0.36 GHz from the junction-array mode placed at 10.78 GHz by Manucharyan's extended model, while the 0–3 coupling, which we compute from that model, is approximately 0.53 GHz. The coupling therefore exceeds the detuning by a factor of 1.5–3.7, placing the transition in a strongly hybridized regime that cannot be captured by the bare-circuit model. Explicit diagonalization of the fluxonium coupled to both array modes confirms that this discrepancy is concentrated near $\Phi_{\text{ext}} = 0$. The dressed f_{01} shifts by -16% there, but by only -0.43% at the flux sweet spot, where the coherence calculations in this study are performed. This residual shift is smaller than the 9.5% spread produced by the paper's own published parameter sets.

Finally, the second-order channel in Eq. (2) is the only physical contribution implemented here rather than inherited from established tooling, so we check it against an independent published evaluation rather than against its defining equation alone. Nguyen *et al.*, using the same expression and their measured $A_\Phi = 1.8\,\mu\Phi_0$, state that the second-order limit exceeds 10 ms for all of their qubits [2]. Evaluated for their nine devices at the same amplitude, our implementation gives $T_\phi^{(2)} = 15$–804 ms, consistent with that statement. This tests the rate expression in the parameter regime of the published devices; it does not independently validate its extrapolation to the much deeper-well designs of Section 6.

5 The Transmon Design Plane

We sweep $E_J \in [1, 80]$, $E_C \in [0.1, 2.0]$ GHz at $n_g = 0$ and $n_g = 0.25$ (800 points each), and refine to a 65×53 grid over $E_J \in [4, 20]$, $E_C \in [0.08, 0.6]$ GHz.

5.1 The confound

Within the transmon regime $E_J/E_C \geq 20$, $\text{corr}(\log T_2, \log f_{01}) = -0.9999$. Under this noise model the coherence landscape over the plane is effectively one-dimensional: E_J and E_C remain independent design variables, but T_2 depends on them almost only through f_{01}. The reason is structural rather than numerical: dielectric loss dominates T_1 everywhere in the regime, and at transmon frequencies $hf_{01} \gg k_B T$, with $\Gamma \propto |\langle 0|\hat{n}|1\rangle|^2 S(\omega_{01})$. Since $|\langle 0|\hat{n}|1\rangle|^2 \propto \sqrt{E_J/E_C}$ and $S \propto E_C$, the rate goes as $\sqrt{E_J E_C} \propto \omega_{01}$—a function of frequency alone. The critical-current rate, set by $E_J\,\partial\omega_{01}/\partial E_J \approx \omega_{01}/2$, is too. Consequently the best sampled T_2 point of the swept plane, 147 µs at $E_J = 3.03$, $E_C = 0.10$ GHz, sits at the corner of the sampled domain—the smallest E_C on the grid and the smallest sampled E_J satisfying $E_J/E_C \geq 20$ (the E_J grid spacing is 2.03 GHz, so the exact boundary $E_J = 2.0$ GHz is not sampled)—at $f_{01} = 1.45$ GHz, below the conventional 3 GHz to 6 GHz transmon operating band [18]. It is a frequency minimum, not a design.

5.2 No interior optimum in this model

Holding $f_{01} = 5.00 \pm 0.05$ GHz and sweeping E_J/E_C from 11.4 to 117.6, a cut that extends below the $E_J/E_C \geq 20$ regime to show the trend, T_2 is flat to 3.2% (42.64 µs to 44.02 µs) while $|\alpha|$ falls by a factor of 7.3 (1353 MHz to 185 MHz). **Charge insensitivity is bought with anharmonicity, and essentially not with coherence at all** (Fig. 2). Lower anharmonicity

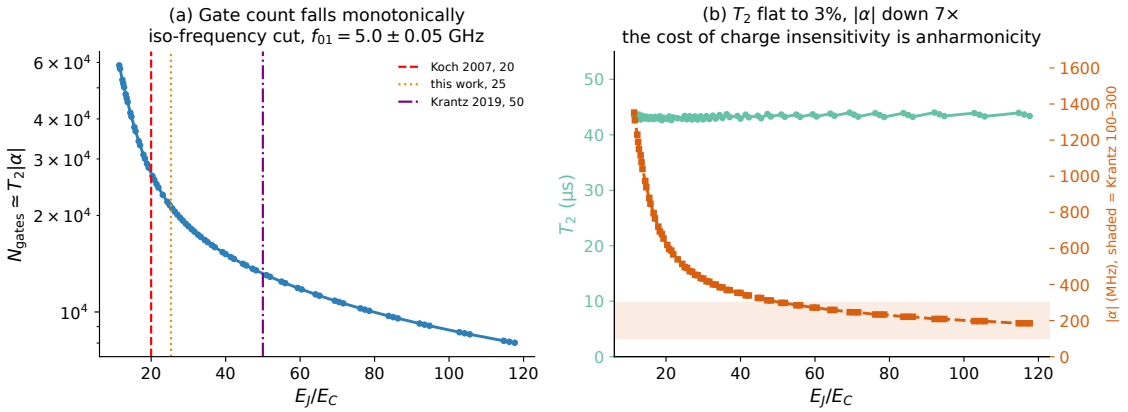


Figure 2: The transmon trade-off at fixed frequency ($f_{01} = 5.00 \pm 0.05$ GHz). (a) The anharmonicity-based gate-count estimate falls monotonically with E_J/E_C; the three vertical lines are candidate regime thresholds, and the optimum sits on whichever one is imposed. (b) T_2 is flat to 3.2% while $|\alpha|$ falls 7.3×; the shaded band is the Krantz 100 MHz to 300 MHz design window. Charge insensitivity costs anharmonicity, not coherence.

tightens the leakage constraint on pulse design, but how much it lengthens a real gate depends on pulse shaping, matrix elements and the error target, as the pulse simulation of Section 8.4 shows.

That the trade is anharmonicity-for-charge-insensitivity is expected from Koch's asymptotics: with $\alpha \simeq -E_C$ and $f_{01} \simeq \sqrt{8E_JE_C}$ held fixed, raising E_J/E_C forces $E_C \propto (E_J/E_C)^{-1/2}$, predicting a factor 3.21 over our swept range against a computed E_C ratio of 3.47. The observed 7.3× fall in $|\alpha|$ is steeper still because $\alpha \simeq -E_C$ is asymptotic and fails at the low end: $|\alpha|/E_C = 2.29$ at $E_J/E_C = 11.4$ against 1.09 at 117.6. The trade-off is therefore harshest exactly in the low-E_J/E_C corner that an unconstrained gate-count metric wants to select, which is an independent reason to keep the design envelope.

Relocating the gate-count optimum under eight regime thresholds spanning $E_J/E_C = 15$ to 60, the optimum tracks every one of them to within 0.24. It sits *on* the constraint, never inside the domain. This is the signature of a constrained optimum, and it means the substantive choice in this part of the design space is the threshold, not the optimization. The correct claim is a monotone trade-off terminated wherever the regime boundary is drawn, and any quoted optimum is meaningful only together with its threshold. We adopt the conventional design envelope of Krantz *et al.* ($E_J/E_C \geq 50$, $|\alpha| \in [100, 300]$ MHz, $f_{01} \in [3, 6]$ GHz) [18] because it is independent of every noise amplitude in the study, giving the operating point $E_J = 13.0$ GHz, $E_C = 0.26$ GHz, $f_{01} = 4.92$ GHz, $T_2 = 44.1$ μs, $|\alpha| = 299$ MHz.

A limitation of the model bears on this choice: `tphi_1_over_f_ng` covers $1/f$ dephasing only, and the broadband charge sensitivity that motivates Krantz's $E_J/E_C \geq 50$ is absent. Left to itself the metric under-penalizes low E_J/E_C and would recommend $|\alpha| \approx 728$ MHz, roughly 2.4× what is typically fabricated. The envelope is a constraint imposed on the model, not a result derived from it.

6 The Fluxonium Design Plane

Over $E_J \in [3, 15]$, $E_L \in [0.2, 1.5]$ GHz at $E_C = 2.5$ GHz and $\Phi/\Phi_0 = 1/2$, the same confound appears: $\mathrm{corr}(\log T_2, \log f_{01}) = -0.9878$. It is looser than for the transmon. About a log-log fit

(slope −0.79), fluxonium T_2 at fixed f_{01} still scatters by 12% rms, and by up to 39% at the worst point, against 0.6% rms for the transmon; frequency is the dominant variable here, not the only one. The unconstrained maximum, 1398 µs, sits at both grid edges at $f_{01} = 75$ MHz, where $\bar{n}_{\rm th} = 3.7$, and 2.3× below the lowest-frequency device in Nguyen *et al.*'s Table I. Constrained to that table's range, 0.17 GHz to 0.83 GHz, the best sampled design gives 719 µs at $E_J = 10.6$, $E_L = 0.20$ GHz ($f_{01} = 0.177$ GHz), itself on the E_L grid edge and at the low end of the band: the confound again.

Here and below, $\bar{n}_{\rm th} = 1/(e^{hf_{01}/k_BT} - 1)$ is the Bose occupation of the bath at the qubit frequency, at the model's $T = 15$ mK. It is not the qubit's own population. The qubit's equilibrium excited-state population, $p_1 = 1/(1 + e^{hf_{01}/k_BT})$, is 44% at 75 MHz, but it is already 24% for the nominal design at 0.363 GHz: thermal population is a matter of degree across this whole regime, and under these conditions passive thermal equilibrium does not provide high-fidelity ground-state initialization for any fluxonium here; active reset, heralding or engineered cooling is needed. Effective qubit temperatures in real devices are often above the bath temperature assumed here.

6.1 An interior optimum, created by second-order dephasing

Extending the scan to $E_J = 45$ GHz reveals a one-dimensional interior maximum of T_2 in E_J within this model, present in eight of the nine (E_C, E_L) slices tested; this is not a joint optimum over all three parameters, and in the ninth ($E_C = 3.5$, $E_L = 0.2$ GHz) T_2 is still rising at the edge of the scan, so its optimum is not located (Fig. 3). It is created by the second-order flux channel: as E_J rises and E_L falls the qubit sinks into a deep double well, f_{01} collapses, and $\partial^2 f_{01}/\partial\Phi^2$ grows by three orders of magnitude across the scan, so $\Gamma_\Phi^{(2)}$ grows with it and eventually overwhelms the improving dielectric and critical-current terms. Located by continuous optimization in E_J at each of the nine (E_C, E_L) slices rather than read off the $\Delta E_J = 2.5$ GHz grid, the optimum moves by at most 0.9 GHz in E_J and 0.8% in T_2, and at it the second-order channel carries 46–47% of the total decoherence rate $1/T_2$ in the eight located slices. That is what an interior optimum of this kind looks like: the channel that turns the curve over is comparable to everything else at the turn. It is an optimum of this model. Its location inherits the uncertainty in the second-order rate convention (Section 10), the loss model and the omitted channels, and it applies to the slices searched. Two of those dependences are large. Under a $\ln^2$ second-order convention the optimum persists but moves up by an order of magnitude in frequency, to 0.18–0.31 GHz in the two slices checked; and the array-mode bound of Section 4 does not extend to these deep-well designs. A first-order-only pipeline shows no turnover over the same range and rises monotonically, reaching 5.7×10^4 µs at the end of the scan (a model output, not an achievable coherence)—which is precisely how an earlier version of this analysis concluded there was no interior optimum, and is a concrete argument for including the second-order channel in any design study that sweeps toward low E_L.

The mechanism is geometric. As E_J rises and E_L falls the potential minima separate and the tunnel splitting that sets f_{01} collapses; the same deepening well sharpens the curvature to which $\Gamma_\Phi^{(2)}$ is proportional by Eq. (2), so a single geometric change both lowers the frequency and raises the second-order rate. The investigated transmon model shows no analogous turnover: its charge dispersion falls exponentially in $\sqrt{E_J/E_C}$ with no turning point.

The optimum nevertheless carries a cost. In the convention of Eq. (2) it occurs at $f_{01} = 5.4$–30 MHz, with $\bar{n}_{\rm th} = 10$–57 and an equilibrium excited-state population of 48–50%, so passive thermal equilibration at 15 mK does not provide high-fidelity ground-state initialization for these designs. That is a cost, not a barrier: Zhang *et al.* operate a 14 MHz fluxonium, initialize it with 97% fidelity and run single-qubit gates with 99.8% average fidelity [19], and Najera-

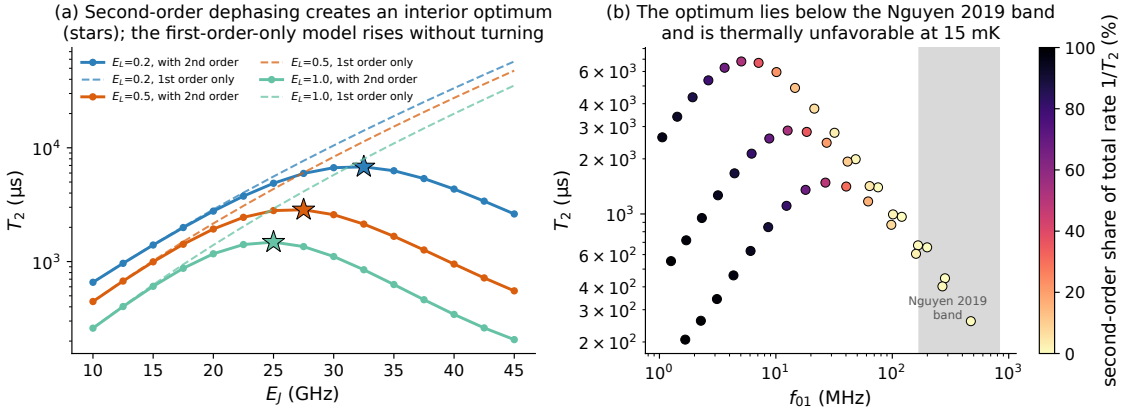


Figure 3: (a) T_2 versus E_J at three values of E_L. Solid curves include second-order flux dephasing and turn over (stars); the dashed first-order-only curves show no turnover over the plotted range, which is how the interior optimum was previously missed. (b) The same points against f_{01}, colored by the second-order share of the total decoherence rate $1/T_2$. The shaded band is the frequency range of Nguyen *et al.*'s devices; every turning point lies below it, at frequencies comparable to operated low-frequency fluxoniums (1.8 and 14 MHz) [19, 20].

Santos *et al.* cool a 1.8 MHz fluxonium to a 97.7% ground-state population [20]. Under the 15 mK equilibrium assumed here the optimum is thermally unfavorable rather than impossible; whether it is worth reaching is a question of initialization overhead, which this model does not include, and these results do not show that the particular designs found here would work experimentally. Past the turnover the frequency correlation reverses sign: within each slice of Fig. 3, $\mathrm{corr}(\log T_2, \log f_{01})$ is +0.98 to +0.99 beyond the optimum, against −0.99 over the in-band plane. The second-order channel adds new structure rather than rescaling the old.

7 Dominant Noise Channels

At a charge-sensitive bias ($n_g = 0.25$) charge noise dominates the transmon below $E_J/E_C \approx 25$ and dielectric relaxation above it. That crossover is computed from our numerical sweep and lands close to the lower edge of the transmon window of Koch *et al.*, $20 \lesssim E_J/E_C \ll 5\times10^4$ [1]. The two are not independent: both reflect the exponential suppression of charge dispersion, and Koch *et al.*'s pulse-duration argument sets only the upper end of their window, through the shrinking anharmonicity. The agreement is therefore a consistency check, and it does not by itself validate the noise amplitudes or the predicted coherence times. Critical-current noise never dominates the transmon anywhere in its 1600 points at literature amplitudes, though it carries about a third of the rate at the operating point. The crossover is noise-model dependent (it spans 20.8–29.9 across literature ranges) and is therefore reported as a model result, not a universal threshold, rather than used as an input.

For the fluxonium the picture is different. Critical-current dephasing dominates at the sweet spot, where it carries 76% of $1/T_2$, and at 358 of the 400 points of the design plane, and first-order flux noise becomes the largest single term only beyond $8.7\times10^{-5}\,\Phi_0$. That makes the fluxonium results sensitive above all to A_{cc} and its convention (Section 3.1). Quasiparticle tunneling is not a dead channel: it is suppressed only *at* the sweet spot, where its 0–1 matrix element vanishes by symmetry, and limits T_1 at most detuned points. This finding is conditional on $x_{qp} = 3\times10^{-6}$, a default roughly 300× above the bound of 10^{-8} that Nguyen *et al.* infer for

their devices; at their density the channel is negligible everywhere. It is one of the most fragile results in the study, after the critical-current dependence above, and is reported as conditional.

8 The Cross-Architecture Comparison

The comparison uses the transmon operating point of Section 5 ($E_J = 13.0\,\mathrm{GHz}$, $E_C = 0.26\,\mathrm{GHz}$) and the nominal fluxonium $E_J = 8.9\,\mathrm{GHz}$, $E_C = 2.5\,\mathrm{GHz}$, $E_L = 0.5\,\mathrm{GHz}$ at half flux ($f_{01} = 0.363\,\mathrm{GHz}$, $T_2 = 368.8\,\mu\mathrm{s}$). The fluxonium is chosen because it is the spectroscopically validated device of Section 4, not the best design in band: the best Table I-band design of Section 6 would raise the raw ratio by a further 1.9×, mostly by lowering the frequency.

8.1 Bias robustness, the one unambiguous asymmetry

Both bias coordinates are periodic with period one, so the width of the window over which T_2 stays within a factor k of its sweet-spot value can be compared as a fraction of one bias period. This is a comparison of normalized sensitivity, not of the physical precision each bias demands: offset charge is not normally controlled at all, which is why insensitivity to it matters, while flux is set by a bias current. The transmon's T_2 varies by 1.0040× across its *entire* charge period—a factor-of-two loss never occurs at any bias. The fluxonium loses a factor of two within $\pm 1.11 \times 10^{-4}\,\Phi_0$ and a factor of ten within $9.96 \times 10^{-4}\,\Phi_0$, located by root-finding on the model rather than read off a grid. The usable bias window is therefore at least 4.5×10^3 times wider for the transmon, a lower bound because the transmon never reaches the twofold loss.

The falloff exponent is not a single number. It approaches the first-order $1/f$ prediction of -1 only in an intermediate window where T_2 is genuinely flux-dephasing limited, reaching -0.929 there, and flattens on both sides—capped by the $2T_1$ ceiling close in, and contaminated by other channels far out.

Intrinsic $1/f$ flux noise does not by itself drive the qubit off the spot: at Nguyen's measured amplitude, in the two-sided convention $S_\Phi(\omega) = 2\pi A_\Phi^2/|\omega|$ used for every dephasing rate here, the variance accumulated over both frequency signs between $f_{\mathrm{lo}} = 1/t$, for an observation time t of one hour, and $f_{\mathrm{hi}} = 1\,\mathrm{GHz}$ is $\sigma_\Phi^2 = 2A_\Phi^2 \ln(f_{\mathrm{hi}}/f_{\mathrm{lo}})$, an RMS wander of $1.4 \times 10^{-5}\,\Phi_0$ (a one-sided reading of the same amplitude would give $9.7 \times 10^{-6}\,\Phi_0$; note that this drift estimate uses $1/t$ as its infrared cutoff, not the ω_{low} of the coherence calculation), a 73× margin to the tenfold-loss detuning. Under that spectrum and those limits, systematic drift and bias calibration may become the more restrictive constraints rather than stochastic noise—which relocates the difficulty rather than removing it, and plausibly makes it grow with qubit count, although a single-qubit model cannot show that. A different spectrum or a longer observation time would change the margin.

This constraint is being engineered away, which we should say plainly. Bénâtre *et al.* hold gradiometric fluxonium qubits at their sweet spot passively, reporting stability of $3 \times 10^{-4}\,\Phi_0$ from target with only a 15% reduction in echo coherence [21]. Their achieved stability lies between our computed twofold- and tenfold-loss detunings (1.1×10^{-4} and $1.0 \times 10^{-3}\,\Phi_0$), and their 15% loss is smaller than our model would give there—unsurprisingly, since their device has different E_L and hence different curvature, and echo suppresses low-frequency drift that free induction does not. We read their result as evidence that the bias-precision cost we quantify is real, understood, and addressable by design rather than by calibration alone. It weakens the scalability pessimism in our conclusion without touching the arithmetic behind it.

Table 2: Fluxonium/transmon ratio by metric. MC = median and 5–95% sample range over $N = 400$ log-uniform draws (fixed seed) of $Q_{\mathrm{cap}} \in [10^5, 10^7]$, $A_\Phi \in [10^{-6}, 10^{-5}]\Phi_0$, $A_{\mathrm{cc}} \in [10^{-7}, 10^{-6}]$ and $A_{n_g} \in [10^{-4}, 10^{-3}]$; the three amplitude ranges extend one decade above the defaults, so the Monte Carlo is noisier than the deterministic column. All Monte Carlo columns use shared draws, the same noise parameters for both architectures in each sample. For the matched-frequency row the deterministic values span the pairs at 0.5, 1 and 2 GHz, and the Monte Carlo is at 1 GHz. With independent draws the matched-frequency median is 0.47× and the fluxonium wins 19%.

Metric	Det.	MC med. [5–95%]	Wins
Raw T_2	8.4	5.1 [3.6, 13.1]	100%
T_2, matched f_{01} (MC at 1 GHz)	0.62–0.82	0.37 [0.25, 0.99]	**5%**
Gates, $1/\lvert\alpha\rvert$	242	148 [105, 379]	100%
Gates, $1/\min(\lvert\alpha\rvert, f_{01})$	10.2	6.2 [4.4, 15.9]	100%
Gates, $1/f_{01}$	0.62	0.38 [0.27, 0.97]	**4%**
Gates, simulated	22	–	–
Bias window, 2× loss	$\geq 4.5 \times 10^3$ wider for the transmon		

8.2 No single ratio

Table 2 gives the comparison under every metric we consider defensible, and Fig. 4 shows how each fares across the noise band.

Two ranges are visible, and they are of comparable size. *Across* metric definitions—a spread, not an uncertainty on one physical quantity—the answer spans roughly 400× (242× against 0.62×, Table 2). *Within* the gate-count proxy alone it spans 393×, depending on the heuristic pulse-duration estimate used. The $|\alpha|$ bound—leakage to $|2\rangle$ only—is the textbook choice, but for a fluxonium at half flux $|\alpha| = 8.7$ GHz is 24× the transition frequency it would drive, so that bound licenses a pulse 24× shorter than one period of the transition itself. The drive-period estimate $1/f_{01}$ gives 0.62×. The three are diagnostic estimates that show how sensitive the answer is to the pulse assumption, not validated limits on gate time. The $1/f_{01}$ estimate ignores leakage entirely: for the transmon it would allow a pulse 16.5× shorter than $1/|\alpha|$, with no evidence that it meets an error target. Nor is one drive period a hard floor: Zhang *et al.* perform their computational fluxonium gates within a single transition period [19].

8.3 The raw advantage is a frequency effect

The two devices differ in frequency by 13.6×. If $T_2 \propto 1/f_{01}$, the frequency ratio alone would produce $\log 13.55 / \log 8.36 = 123\%$ of the log-ratio: more than the whole raw advantage, so at fixed frequency the fluxonium would be the less coherent of the two. This is a scaling estimate under that law, not a decomposition of causes, and the law describes the transmon better than the fluxonium. Dielectric loss dominates the transmon and critical-current dephasing the fluxonium. For the transmon the rate scales with ω_{01} and $T_2 \propto f_{01}^{-1.00}$ is obtained from a fit to the modeled design plane; for the fluxonium the thermal factor, the matrix elements and the dominant critical-current channel break a simple $\Gamma_1 \propto \omega_{01}$ law, and the scaling, $T_2 \propto f_{01}^{-0.79}$ ($r = -0.988$), is an empirical property of the model rather than a derived one. Because the flux sweet spot *is* a frequency minimum, the fluxonium is structurally advantaged on any uncontrolled T_2 comparison, and reporting one credits the architecture with what the frequency

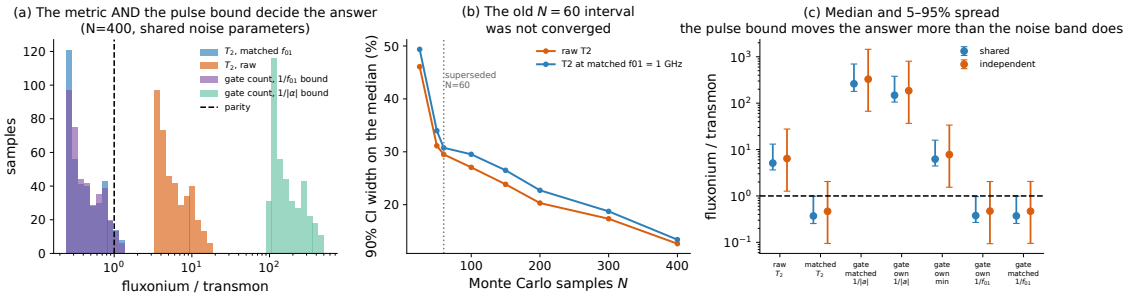


Figure 4: Robustness of the Q3 answer to the noise amplitudes. (a) Distribution of the fluxonium/transmon ratio under four metrics, $N = 400$ shared-parameter draws; the metric and the pulse bound each move the answer by more than the noise band does. (b) Width of the 90% bootstrap CI on the median against sample size. It narrows with N; at $N = 400$ the matched-frequency median is 0.37× with CI 0.355–0.404, against a much wider interval at the $N = 60$ of an earlier version. Panels (a) and (c) show the spread under the priors; (b) shows the precision of the estimated median. (c) Median and 5–95% spread per metric under shared versus independent sampling.

did.

Controlling for it, the deterministic ratio at matched f_{01} is already below one, 0.62–0.82× at 0.5, 1 and 2 GHz. Propagated across the literature noise band at matched $f_{01} = 1$ GHz, with shared draws, the median is 0.37× and the fluxonium wins in only 4.8% of samples (19% with independent draws). The 90% bootstrap interval on that median is 0.355–0.404. These are draws of the noise parameters under our priors, not a population of fabricated devices. **Under these noise priors, and at matched frequencies where the transmon is not a practical design, the transmon comes out ahead in 95% of samples.** This is the most important negative result here: in this model no coherence advantage remains once frequency is controlled.

8.4 Estimating gate durations by simulation

The gate-count range is wide enough to make the metric almost uninformative, so we estimate it by simulation rather than by argument. We integrate the lab-frame Schrödinger equation for each architecture with no rotating-wave approximation, using a six-level truncation of the `scqubits` eigenbasis and a charge drive, and numerically optimize the pulse amplitude, DRAG coefficient [22] and Stark-shifted drive frequency at every duration; the average gate error is evaluated on the computational subspace so that leakage is counted; Fig. 5 shows the resulting error curves.

This exposes an effect that the anharmonicity argument misses entirely. The fluxonium's $|\alpha|/f_{01} = 24$ looks decisive, but its drive matrix elements are unfavourable: at half flux $|\langle 1|\hat{n}|2\rangle|/|\langle 0|\hat{n}|1\rangle| = 10.2$, against 1.37 for the transmon. The drive couples ten times more strongly to the leakage transition than to the qubit transition. That ratio is an important factor the anharmonicity alone does not capture; leakage also depends on the pulse spectrum, detuning, drive strength and higher levels, which is why we simulate rather than combine the two numbers by hand.

Table 3 reports the shortest tested duration at which the calibrated gate reaches an average error below 10^{-3} for each architecture, and the resulting gate count. *These durations are upper bounds*: a three-parameter calibration over a fixed Gaussian shape is far from optimal control, and more flexible optimal-control approaches, such as GRAPE or CRAB over a free pulse shape,

Table 3: Shortest tested duration at which the calibrated X gate reaches an average error below 10^{-3}, from the lab-frame simulation, and the resulting gate count $N = T_2/T_{\text{gate}}$, a coherence-limited gate-time budget rather than a number of gates that could be run in sequence: gate errors accumulate, and the calibration omits decoherence during the pulse. A passing duration shows that a gate that short is achievable within the simulated model (charge drive, one pulse family). The listed intervals bracket the shortest passing durations found under the explored optimization settings; they are not rigorous bounds on globally optimal gate times. A stronger optimizer gives 20×.

	T_{gate} (ns)	T_2 (μs)	N
Transmon	8.0 (6.0–8.0)	44.1	5,513
Fluxonium	3.0 (2.5–3.0)	368.8	122,928
Simulated ratio			22×
bracket from grid + optimizer			17–27×
for comparison, $1/\lvert\alpha\rvert$ estimate			242×
for comparison, $1/f_{01}$ estimate			0.62×

could shorten one or both gates, potentially by different amounts. The bracket also leaves out modeling choices that a different simulation could make differently: we drive through the charge operator, whereas low-frequency fluxoniums are often controlled through flux [19]; we use a single pulse family, omit decoherence during the gate, and fix the error target at 10^{-3}. The result is a single-model estimate that has not been checked against an independent code or a measured gate. The point is not the precise value but that, within this model, the simulated ratio falls well between the two naive estimates.

We tested the numerics at the durations that set T_{gate}. The pass/fail classification at the tested durations is stable across truncations after recalibration, although individual optimized pulses are not: a pulse calibrated on six levels does not carry over unchanged (the 3 ns fluxonium pulse has error 4×10^{-2} when replayed on ten levels), but recalibrating on eight or ten levels changes no pass/fail verdict at any tested duration, and tighter ODE tolerances reproduce every error. The optimizer is not converged. With seven starting points and 400 evaluations instead of three and 140, both gates shorten, to 6.0 ns (transmon) and 2.5 ns (fluxonium), on six or ten levels, and the ratio becomes 20×, inside the bracket; shorter durations were not retested. The fluxonium gate also needs a Rabi frequency of ≈0.35 GHz to 0.40 GHz, comparable to its own f_{01}; whether a charge drive that strong is practical is not assessed here.

Within the investigated charge-drive model, the fluxonium yields a larger coherence-to-gate-duration proxy than the transmon, by approximately 17–27× over the explored numerical settings, rather than the ∼240× of the anharmonicity estimate or the ∼0.6× of the drive-period estimate, both heuristic pulse-time estimates rather than demonstrated minimum durations. Within this model neither extreme is reached. The quoted bracket covers only the duration grid and optimizer convergence, not the modeling choices listed above, so the true uncertainty is larger than the bracket.

9 Calibration Against Measurement

Run at the parameters of each of the nine devices in the published Table I of Nguyen *et al.*, the model is optimistic by a median factor of 2.0× on T_1 and 1.5× on T_2, worst case 3.2× (device F). The gap is not uniform, so it is not a single missing constant: the T_2 ratio runs

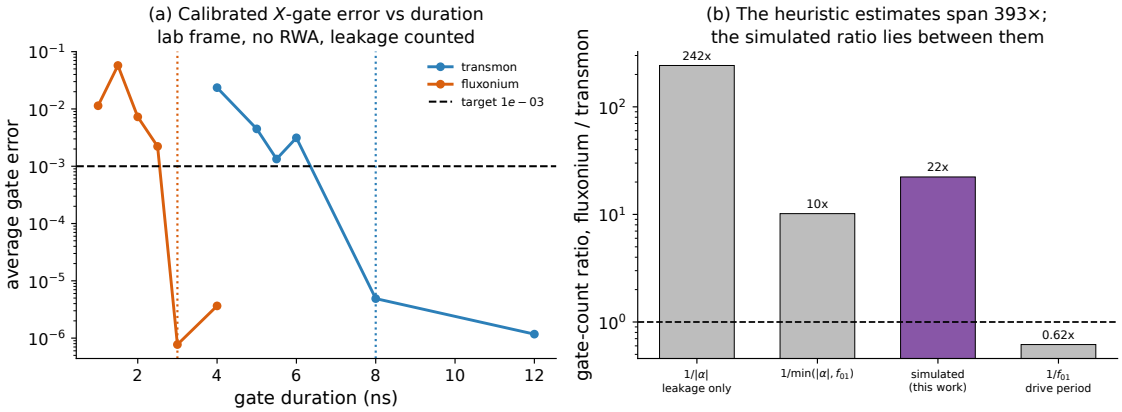


Figure 5: (a) Simulated coherent average X-gate error (leakage included, decoherence omitted) against gate duration for both architectures, from the lab-frame simulation with amplitude, DRAG and Stark detuning optimized at each point; dotted lines mark the shortest tested duration meeting the 10^{-3} target. Non-monotonicity at the shortest durations is optimizer noise and is not smoothed. (b) The gate-count ratio under each pulse-length estimate. The two heuristic pulse-time estimates differ by 393×; the simulation (purple) falls between them.

from 0.57× to 3.2×, and device I, the device nearest our nominal design in frequency (0.395 versus 0.363 GHz), is the one on which the model is *pessimistic* on both, by 1.25× on T_1 and 1.75× on T_2. Under the absolute reading of A_{cc} the T_2 median would be 2.6× (Section 3.1). This calibration covers the fluxonium only. We have not calibrated the transmon model against measured devices, so the cross-architecture ratios of the previous section carry a transmon-side model bias that we do not quantify.

Possible contributors include the flat $Q_{cap} = 10^6$ against their measured $\tan\delta_C = (2.0–3.6) \times 10^{-6}$ at 6 GHz; the absence of Purcell decay, photon shot noise and inductive loss; the fact that their T_2 is a Hahn echo while this model computes free-induction decay, so the two are not like for like (because echo generally suppresses low-frequency dephasing, an echo version of the model could give a longer T_2 and change the discrepancy; we do not quantify that protocol dependence); and a temperature mismatch, since we run at the 15 mK default while they state 20 mK. We have not quantified these contributions separately, so the list is not a ranking.

The comparison also has a floor set by our own model rather than by the devices. Absolute T_2 spans 47 µs to 1163 µs under single-decade moves of the noise amplitudes, so a factor of 1.5 is well inside the band the model itself admits; the calibration should be read as establishing the right order of magnitude, not as a residual to be explained away, and, because the protocols differ, not as a precise estimate of omitted loss. Device performance and loss parameters also differ across fabrication generations: Somoroff *et al.* report $T_2^* = 1.48$ ms [6], well beyond the devices we calibrate against, so a calibration to Nguyen's 2019 devices should not be assumed to predict later hardware without re-evaluating the physical parameters.

We stress that the published table must be used. The arXiv v1 preprint lists eight devices and omits device I entirely—the paper's headline device, and the one closest to the design regime studied here.

10 Limitations

The model omits Purcell decay, photon shot noise, inductive loss in the junction array, TLS spectral structure, crosstalk and any lattice; qubits are treated in isolation. The superinductance is ideal, so array-mode hybridization is absent, though Section 4 bounds that omission at −0.43% in f_{01} at the flux sweet spot for the nominal and Manucharyan devices; it is not bounded for the deep-well designs of the extended scan. Second-order dephasing uses a convention with no logarithmic factor; a $\ln^2$ convention would change sweet-spot T_2 by ∼16% instead of 0.2% and would move, though not remove, the interior optimum: in two slices we checked it rises by an order of magnitude in frequency, to 0.18–0.31 GHz, inside the range of Nguyen *et al.*'s devices. The location of the optimum is therefore the least robust number in Section 6. Absolute T_2 spans 47 µs to 1163 µs under single-decade moves of the noise amplitudes and should never be quoted without a band. The Monte Carlo varies amplitudes only; fabrication spread in E_J, E_C, E_L is not propagated, and the log-uniform priors are a choice. The gate-count metric is a heuristic, not a literature formula: $N = T_2/T_{\text{gate}}$ is a ratio of timescales, not an achievable circuit depth.

Several checks that would strengthen these results have not been done, and the conclusions are stated for this model only because of them. No result has been reproduced with an independent implementation; everything comes from one code base built on `scqubits`. The interior fluxonium optima are located by continuous optimization, but the transmon optimum and the in-band fluxonium optimum are read from sampled grids rather than solved along the constraint boundaries. The second-order curvature is computed by a single method, Richardson-extrapolated finite differences. The Monte Carlo uses one fixed seed and priors of our choosing. The pulse simulation uses one propagator and one pulse family, its optimizer is not converged, and shorter durations were not retested with the stronger optimizer. Noise-model parity has not been tested: adding quasiparticle loss to the transmon at the fluxonium's density, or inductive loss to the fluxonium, could change the matched-frequency ranking, which Section 3.3 shows already depends on the loss model, and Section 2.3 on the choice of matched transmon.

11 Conclusion

Neither architecture has a coherence sweet spot in design space that comes without a cost. Over the swept design planes the modeled T_2 is, to better than $|r| = 0.98$, largely a function of the transition frequency (to 0.6% rms for the transmon and 12% for the fluxonium), so a raw T_2 maximum locates the edge of the swept domain or the point where a frequency constraint was drawn. The extended fluxonium scan departs from this: past its interior optimum, lowering the frequency further makes T_2 worse. The transmon is a monotone trade-off in which charge insensitivity is paid for in anharmonicity, once, at design time. Within this model the fluxonium has an interior optimum, visible only once second-order flux dephasing is included, but in the convention of Eq. (2) it sits at 5.4–30 MHz, where passive thermal equilibration at 15 mK does not give high-fidelity initialization, and its location depends strongly on the second-order rate convention; without the overhead of active initialization it too is a trade-off terminated by a frequency floor.

For Q3, the honest output is a metric family, not a number. The fluxonium's raw 8.4× coherence advantage is, under $T_2 \propto 1/f_{01}$ scaling, more than accounted for by frequency; at matched frequency the transmon is ahead deterministically at all three frequencies tested, under both loss models (0.62–0.82× with the flat Q_{cap}), and in 95% of noise draws at 1 GHz (median 0.37×). These fluxonium numbers rest mainly on the critical-current amplitude, the

least constrained input in the study: read as `scqubits` applies it, the matched ratios would be 1.07–2.16× instead. Under the tested charge-drive pulse model the fluxonium has a larger coherence-to-gate-duration proxy, by roughly 17–27× across the explored numerical settings. Its price is operating-point precision: a normalized bias window at least 4.5×10^3 times narrower than the transmon's, which must hold during operation, whether by active control, passive flux locking [21] or other architectural means, rather than being fixed once at design time. Whether that trade is worth making depends on qubit count and calibration budget, which this single-qubit model does not quantify.

More generally, we suggest that architecture comparisons in this field should state the metric, the control variable and the uncertainty band together. A single ratio between architectures evaluated at different operating points cannot, by itself, isolate an architecture-dependent advantage.

12 Outlook

Two extensions would sharpen the results most. The gate-count metric is currently estimated with a coarse calibration in a single model; optimal control over a free pulse shape (GRAPE or CRAB), a flux drive, and a check against an independent code or a measured gate would tighten the 17–27× bracket and, more interestingly, would test whether the fluxonium's unfavourable leakage matrix element can be shaped around—the anharmonicity is genuinely there, and a pulse that exploits it without driving $|1\rangle \to |2\rangle$ would move the number back toward the leakage-only bound. Second, passive thermal equilibration at 15 mK does not initialize the interior fluxonium optimum we find with high fidelity, but low-frequency fluxoniums have been operated with active initialization [19, 20]; modeling that initialization, and its cost in time and fidelity, would establish whether the optimum is a usable design point.

A broader point follows from the metric sensitivity. If a single architecture comparison can span more than two orders of magnitude depending on choices that are all defensible, then comparisons reported as single ratios carry less information than they appear to. We would encourage the practice McRae *et al.* [7] advocate for experimental coherence numbers—report the conditions with the number—to be extended to modeled design comparisons: state the control variable, state the figure of merit, and give the spread.

Code and data availability

All code, generated data and figures are openly available at
https://github.com/ishitabera/quantum-coherence-sweet-spots-transmon-fluxonium.
The version that produced every number in this paper (release v1.0) is archived at Zenodo, doi:10.5281/zenodo.23988521.

The results are produced by five notebooks run in order from a single pipeline, plus three standalone check scripts in `results/` for the robustness, pulse-convergence and matched-transmon checks; none reimplements the physics, and the Monte Carlo uses a fixed random seed. Each sweep writes its raw data before any figure code executes, together with a metadata record of the grid, the noise amplitudes, the software versions and the git revision. The headline values are bound to the file and filter that produced them by an automated provenance ledger of 53 entries, which recomputes and asserts each one on every run and fails rather than warns if a value drifts from its artifact. A test suite of 35 checks pins the conventions and the validated anchors, including the published numbers used for calibration.

Acknowledgments

Use of generative AI This work was carried out by the author together with a large language model (Claude, Anthropic), which contributed to the analysis code and the numerical analysis. The author wrote the manuscript with its assistance, reviewed it, double-checked the physics and the numerical results, and is responsible for all claims.

Funding This work was carried out as a self-directed project and received no external funding.

References

[1] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin and R. J. Schoelkopf, *Charge-insensitive qubit design derived from the Cooper pair box*, Phys. Rev. A **76**(4), 042319 (2007), doi:10.1103/PhysRevA.76.042319.

[2] L. B. Nguyen, Y.-H. Lin, A. Somoroff, R. Mencia, N. Grabon and V. E. Manucharyan, *High-coherence fluxonium qubit*, Phys. Rev. X **9**(4), 041041 (2019), doi:10.1103/PhysRevX.9.041041.

[3] C. Wang, C. Axline, Y. Y. Gao, T. Brecht, Y. Chu, L. Frunzio, M. H. Devoret and R. J. Schoelkopf, *Surface participation and dielectric loss in superconducting qubits*, Appl. Phys. Lett. **107**(16), 162601 (2015), doi:10.1063/1.4934486.

[4] M. Papič, J. Tuorila, A. Auer, I. de Vega and A. Hosseinkhani, *Charge-parity switching effects and optimisation of transmon-qubit design parameters*, npj Quantum Inf. **10**, 69 (2024), doi:10.1038/s41534-024-00860-7.

[5] S. S. Ram, M. Molli, V. M. Katukuri and B. C. Mummaneni, *Optimizing superconducting qubit performance: A theoretical framework for design, analysis, and calibration*, arXiv:2501.17825 (2025), Preprint.

[6] A. Somoroff, Q. Ficheux, R. A. Mencia, H. Xiong, R. Kuzmin and V. E. Manucharyan, *Millisecond coherence in a superconducting qubit*, Phys. Rev. Lett. **130**(26), 267001 (2023), doi:10.1103/PhysRevLett.130.267001.

[7] C. R. H. McRae, G. M. Stiehl, H. Wang, S.-X. Lin, S. A. Caldwell, D. P. Pappas, J. Mutus and J. Combes, *Perspective: Reproducible coherence characterization of superconducting quantum devices*, Appl. Phys. Lett. **119**(10), 100501 (2021), doi:10.1063/5.0060370.

[8] F. Lange, L. Heunisch, H. Fehske, D. P. DiVincenzo and M. J. Hartmann, *Cross-talk in superconducting qubit lattices with tunable couplers—comparing transmon and fluxonium architectures*, Quantum Sci. Technol. **11**(1), 015020 (2026), doi:10.1088/2058-9565/ae2358.

[9] G. Catelani, R. J. Schoelkopf, M. H. Devoret and L. I. Glazman, *Relaxation and frequency shifts induced by quasiparticles in superconducting qubits*, Phys. Rev. B **84**(6), 064517 (2011), doi:10.1103/PhysRevB.84.064517.

[10] P. Groszkowski and J. Koch, *scqubits: a Python package for superconducting qubits*, Quantum **5**, 583 (2021), doi:10.22331/q-2021-11-17-583.

[11] D. J. Van Harlingen, T. L. Robertson, B. L. T. Plourde, P. A. Reichardt, T. A. Crane and J. Clarke, *Decoherence in Josephson-junction qubits due to critical-current fluctuations*, Phys. Rev. B **70**(6), 064517 (2004), doi:10.1103/PhysRevB.70.064517.

[12] J. Eroms, L. C. van Schaarenburg, E. F. C. Driessen, J. H. Plantenberg, C. M. Huizinga, R. N. Schouten, A. H. Verbruggen, C. J. P. M. Harmans and J. E. Mooij, *Low-frequency noise in Josephson junctions for superconducting qubits*, Appl. Phys. Lett. **89**(12), 122516 (2006), doi:10.1063/1.2357010.

[13] C. D. Nugroho, V. Orlyanchik and D. J. Van Harlingen, *Low frequency resistance and critical current fluctuations in Al-based Josephson junctions*, Appl. Phys. Lett. **102**(14), 142602 (2013), doi:10.1063/1.4801521.

[14] F. Yan, S. Gustavsson, A. Kamal, J. Birenbaum, A. P. Sears, D. Hover *et al.*, *The flux qubit revisited to enhance coherence and reproducibility*, Nat. Commun. **7**, 12964 (2016), doi:10.1038/ncomms12964.

[15] G. Ithier, E. Collin, P. Joyez, P. J. Meeson, D. Vion, D. Esteve, F. Chiarello, A. Shnirman, Y. Makhlin, J. Schriefl and G. Schön, *Decoherence in a superconducting quantum bit circuit*, Phys. Rev. B **72**(13), 134519 (2005), doi:10.1103/PhysRevB.72.134519.

[16] W. C. Smith, A. Kou, X. Xiao, U. Vool and M. H. Devoret, *Superconducting circuit protected by two-Cooper-pair tunneling*, npj Quantum Inf. **6**, 8 (2020), doi:10.1038/s41534-019-0231-2.

[17] V. E. Manucharyan, J. Koch, L. I. Glazman and M. H. Devoret, *Fluxonium: Single Cooper-pair circuit free of charge offsets*, Science **326**(5949), 113 (2009), doi:10.1126/science.1175552.

[18] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson and W. D. Oliver, *A quantum engineer's guide to superconducting qubits*, Appl. Phys. Rev. **6**(2), 021318 (2019), doi:10.1063/1.5089550.

[19] H. Zhang, S. Chakram, T. Roy, N. Earnest, Y. Lu, Z. Huang, D. K. Weiss, J. Koch and D. I. Schuster, *Universal fast-flux control of a coherent, low-frequency qubit*, Phys. Rev. X **11**(1), 011010 (2021), doi:10.1103/PhysRevX.11.011010.

[20] B.-L. Najera-Santos, R. Rousseau, K. Gerashchenko, H. Patange, A. Riva, M. Villiers, T. Briant, P.-F. Cohadon, A. Heidmann, J. Palomo, M. Rosticher, H. le Sueur *et al.*, *High-sensitivity ac-charge detection with a MHz-frequency fluxonium qubit*, Phys. Rev. X **14**(1), 011007 (2024), doi:10.1103/PhysRevX.14.011007.

[21] D. Bénâtre, M. Féchant, N. Zapata, N. Gosling, P. Paluch, T. Reisinger and I. M. Pop, *Simultaneous sweet-spot locking of gradiometric fluxonium qubits*, Phys. Rev. Applied **24**, 054031 (2025), doi:10.1103/gjks-ctvm.

[22] F. Motzoi, J. M. Gambetta, P. Rebentrost and F. K. Wilhelm, *Simple pulses for elimination of leakage in weakly nonlinear qubits*, Phys. Rev. Lett. **103**(11), 110501 (2009), doi:10.1103/PhysRevLett.103.110501.